Product Specification

产品交付规格书

Prismatic LFP Cells

方形铝壳锂离子电池

Model: A41-V1

型号：A41-V1

Drafted 编制	Product Design Checked by 产品设计审核	Quality Checked by 品质审核	Sales Checked by 销售审核	Approved by 批准
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Customer Recipient 客户接收栏

Company:
公司名称:
Approved by:
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Date:
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EVE Power Co., Ltd

湖北亿纬动力有限公司

Customer Requirements

EVE Power Co., Ltd. Requires customer to provide specific requirements and communicates with EVE Power Co., Ltd. In advance. If special applications and operation conditions are incons istent with the description of this specification, EVE Power Co., Ltd. May design and manufactur e products according to customer's inputs.

客户要求

要求客户写出他们的需求信息并提前与湖北亿纬动力公司沟通。如果客户有一些特别的应用或者操作条件不同于此文件中所描述的，湖北亿纬动力公司可以根据客户的特别要求进行产品的设计和生产。

No. 序号	Special Requirements 特殊要求	Standards 标准
1	/	/
2	/	/
3	/	/
4	/	/
5	/	/

Customer Code 客户代码：__________ Signature 签字：__________ Date 日期：__________

Model 型号	A41-V1	Specification 规格书编号	PBRI-A41-V1-D06-01	Version 版本	B

Revision History
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A	2023.10.30	First issue 新版发行	Jin Tan 谭津
B	2024.2.20	Version update 版本更新	Jin Tan 谭津

目录

Term Definition

术语定义

Term 术语	Definition 定义
Product 产品	Refers to 106Ah rechargeable lithium-ion cell with prismatic aluminum shell manu-factured by EVE Power Co., Ltd. (EVE) in this specification. 本规格书中的"产品"是指湖北亿纬动力有限公司生产的 106Ah 可充电方形铝壳锂离子电池。
Customer 客户	Refers to the buyer in the product sales contract signed with EVE. 指《湖北亿纬动力有限公司产品销售合同》中的买方。
Environment temperature 环境温度	The ambient temperature where the cell is located. 电池所处的周围环境温度。
Cell temperature 电池温度	The temperature measured by temperature sensor installed at the center of cell surf-ace. The selection of temperature sensor and measuring line shall be jointly by EVE and the customer. 由接入电池表面中心的温度传感器测量的电池的温度，温度传感器和测量线路的选择由 EVE 和客户共同商定 。
Charging/Discharging Rate 充电/放电倍率	The ratio between the charging or discharging current and the capacity which mea-sured by the battery management system for many times. For example, when the cell capacity is 106 Ah and the charging or discharging current is 35.3A, the char-ging or discharging rate is 1/3C. When the cell capacity drops to 100 Ah and the charging or discharging current is 33.3A, the charging or discharging rate is 1/3C. 充/放电电流与电池管理系统多次测量的电池的容量的比率。例如，电池容量为 106Ah，当充电或放电电流为 35.3A 时，则充电或放电倍率为 1/3C。当电池容量跌落为 100Ah，充电电流为 33.3A 时，则充电或放电倍率为 1/3C。
State of charge 荷电状态	Under unloaded conditions, the ration of the cell capacity state which measured in ampere-hour or watt-hour to the rated capacity. The abbreviation is expressed by SOC. For example, if the capacity is 106 Ah which considered as 100% SOC, the capacity is 0 Ah, considered as 0% SOC. 在无负载的情况下，以安培小时或者以瓦特小时为单位计量的电池容量状态与额定容量的比值。如：若将容量为 106Ah 的状态视为 100%SOC，则容量为 0Ah 时，SOC 为 0%。
Cycle 循环	The cell is charged and discharged in a cycle according to the prescribed charging and discharging standards. The cycle includes short-term normal charging or a combination of

	regenerative charging and discharging processes. In the charging process, sometimes there is only normal charging and no re-generative charging. The discharge can be formed by combining some partial discharges. 电池按规定的充放标准充放一次为一个循环。循环包括短时的正常充电或者再生充电和放电过程的组合，在充电过程中有时只有正常充电而无再生充电的情况。放电可以由一些部分放电组合在一起形成。
Standard charge 标准充电	The charging mode described in 3.5 of this specification. 本规格书第 3.5 条所述的充电模式。
Standard discharge 标准放电	The discharge mode described in 3.6 of this specification. 本规格书第 3.6 条所述的放电模式。
Open circuit voltage 开路电压（OCV）	The voltage of the cell measured when unloaded or circuit is connected. The abbr -eviation is expressed by OCV. 没有接入任何负载和电路时测得的电池的电压，缩写用 OCV 表示。
DC resistance 直流电阻（DCR）	The ratio of the voltage changes of the cell to the corresponding current change under working conditions, the abbreviation is DCR. 工作条件下电池的电压变化与相应的电流变化之比，缩写用 DCR 表示。
Module 模组	Lithium-ion batteries combined in series and parallel, intermediate products formed between single cell cells and PACK which are integrated with cell monitoring and management devices. 锂离子电芯经串并联方式组合，加装单体电池监控与管理装置后形成的电芯与 pack 的中间产品。
Pulse current 脉冲电流	The current or voltage pulses that appear periodically are called pulse currents. The pulse currents appear either in the same direction or in alternating positive and negative directions. 以周期重复出现的电流或电压脉冲称为脉冲电流，脉冲电流或是以同一方向出现，或是以正、负交替变换方向出现。
Compression force 压缩力	When the module is assembled, the battery bears the force perpendicular to the battery stacking direction. 模组组装时，电池可承受压缩力的安全边界。
The unit of measurement 测量单位	“V” (Volt) 伏特，Voltage 电压单位 “A” (Ampere) 安培，Current 电流单位 “Ah” (Ampere-Hour)安培-小时，Capacity 负荷单位 “Wh” (Watt-Hour)瓦特-小时，Energy 能量单位 “Ω” (Ohm) 欧姆，Resistance 电阻单位 “mΩ”(Milliohm) 毫欧姆，Resistance 电阻单位

	"°C" (degree Celsius) 摄氏度，Temperature 温度单位 "mm" (millimeter) 毫米，length 长度单位 "s" (second) 秒， Time 时间单位 "Hz" (Hertz) 赫兹， Frequency 频率单位 "W"（Watt）瓦特，Power 功率单位

1. Fundamental Information 基本信息

1.1. Scope of Application 适用范围

This standard describes the product types, basic performance, test methods and precautions of square aluminum shell lithium iron phosphate batteries manufactured by EVE Power Co., Ltd.

本标准描述了湖北亿纬动力有限公司生产的方形铝壳磷酸铁锂电池的产品类型、基本性能、测试方法和注意事项。

1.2. Product Type 产品类型

Prismatic LFP Cell with Aluminum Shell

方形铝壳锂离子电池

1.3. Product Model 产品名称

A41-V1

2. Cell Specification 电池规格参数

2.1. Fundamental Parameters 电池基本参数

The basic parameters of the battery are as follows.

电池的基本参数如下表

Table 1 Basic parameters of cell

表 1 电池基本参数

Items 项目	Standards 标准	Remarks 备注
Min. Capacity 最小容量	106 Ah	1/3C, 25°C±2°C, 2.50~3.65V
Min. Energy 最小能量	341.3 Wh	1/3C, 25°C±2°C, 2.50~3.65V
Initial IR 初始内阻	≤0.40 mΩ	AC, 1kHz, 20%~30%SOC
Nominal Voltage 标称电压	3.22 V	1/3C, 2.50~3.65V
Weight 电池重量	1880±30 g	/
Charging Cut-off Voltage 充电限制电压（U_{max}）	3.65 V	/
Discharging Cut-off Voltage 放电截止电压（U_{min}）	2.50 V(T>20°C) 2.00 V(-30°C≤T≤20°C)	/
Standard Charging Current 标准充电电流	35.3 A	1/3C

Standard Discharging Current 标准放电电流		35.3 A	1/3C
Cycling Performance 循环性能	25°C cycle 25°C循环	2000 Cycles	Under 300Kgf ± 20Kgf initial compression force, FC/1C, 2.5V~3.65V, capacity retention≥80%, or follow the EVE recommended cycling method. 300±20Kgf 初始压缩力下，FC/1C，2.5V~3.65V，循环至容量保持率为80%，或者按照EVE提供的循环方式进行
	45°C cycle 45°C循环	1200 Cycles	
Operation Temperature 工作温度	Charging Temperature 充电温度	-20~55 °C	/
	Discharging Temperature 放电温度	-30~55 °C	/
Storage Temperature 存储温度	1 year 1年	0~25 °C	Delivery SOC State 出货SOC状态
	3 months 3个月	-40~45 °C	
	1 month 1个月	-40~60 °C	

2.2. Product Parameters 产品规格

2.2.1. Dimension and Weight 尺寸、重量指标

Table 2 Cell size and weight parameters
表2 尺寸、重量指标

No.	Items 项目		Standards 标准	Testing Methods 测试方法
1	Dimension 尺寸	Terminal Height 高度 (L_1) (含极柱)	118.5±0.5 mm	3.8.1
		Can-top Height 高度 (L_2) (不含极柱)	115.7±0.5 mm	
		Length 宽度（H）	148.6±0.5 mm	
		Thickness 厚度（T）	52.2±0.3 mm	
2	Weight 重量	Weight (Including blue film, can-top film) 重量(含蓝膜、顶贴片)	1880±30 g	3.8.2

2.2.2. Electrical Performance Parameters 电性能指标

Table 3 Cell electrical performance parameters
表 3 电性能指标

No.	Items 项目		Standards 标准	Testing Methods 测试方法
1	Nominal Capacity 容量	1/3C Capacity 1/3C 容量	≥106 Ah	3.8.3.1
		1C Capacity 1C 容量	≥104 Ah	3.8.3.2
2	Nominal Energy 能量	1/3C Energy 1/3C 能量	≥341.3 Wh	3.8.3.1
		1C Energy 1C 能量	≥329.68 Wh	3.8.3.2
3	Discharge Performance 放电性能	-20°C Capacity retention -20°C容量保持率	≥80%	3.8.3.3
		0°C Capacity retention 0°C容量保持率	≥85%	3.8.3.4
		35°C Capacity retention 35°C容量保持率	≥100%	3.8.3.5
4	DCR	25°C, 50%SOC, 2C, 10sec，Dch	≤0.9mΩ	3.8.3.6
5	Cycle 循环	25°C fast charge cycle 25°C快充循环	2000 cycles, capacity/nominal capacity≥80% 2000 周，容量保持率≥80%	3.8.3.7
		45°C fast charge cycle 45°C快充循环	1200 cycles, capacity/nominal capacity≥80% 1200 周，容量保持率≥80%	3.8.3.8
6	Storage 存储	25°C, 30days, fresh battery, 100 % SOC 25°C, 30 天，新鲜电池, 100%SOC	Capacity retention≥97% 容量保持率≥97%	3.8.3.9
			Capacity recovery≥98% 容量恢复率≥98%	
		45°C, 30 days, fresh battery, 100 % SOC 45°C, 30 天，新鲜电池, 100%SOC	Capacity retention≥96% 容量保持率≥96% Capacity recovery≥97% 容量恢复率≥97%	3.8.3.10

2.2.3 Safety Performance parameters 安全性能指标

Table 4 Cell safety performance parameters
表 4 安全性能指标

No.	Items 项目	Standards 技术标准	Testing Methods 测试方法章节

1	Over Discharge 过放电	No fire, No explosion 不爆炸、不起火	3.8.4.1
2	Over Charge 过充电	No fire, No explosion 不爆炸、不起火	3.8.4.2
3	Extrusion Test 挤压测试	No fire, No explosion 不爆炸、不起火	3.8.4.3
4	External Short-circuit 短路测试	No fire, No explosion 不爆炸、不起火	3.8.4.4
5	Heating 加热测试	No fire, No explosion 不爆炸、不起火	3.8.4.5
6	Temperature Cycle 温度循环	No fire, No explosion 不爆炸、不起火	3.8.4.6

2.3. Cell Drawing 电池图纸

See Appendix 7

见附图 7。

2.4. Out Appearance 外观

The cell should have none of obvious scratches, cracks, rust stains, discoloration, or electrolyte leakage, which have any defects that affect the commercial value of the cell.

电池应无明显擦伤、裂痕、锈渍、变色或电解液泄漏这类对电池商用价值有影响的缺陷。

3. Testing Conditions 试验条件

3.1. Environmental Conditions 环境条件

Unless otherwise specified, the test should be carried out in an environment with a temperature of 25°C ± 2°C, a relative humidity of 15%-90% RH, and an atmospheric pressure of 86 kPa to 106 kPa. The ambient temperature mentioned in this specification refers to 25°C±2°C.

除另有规定外，试验应在温度为 25°C±2°C，相对湿度 15%~90%，大气压力为 86 kPa~106 kPa 的环境中进行。本规格书所提到的室温，是指 25°C±2°C。

3.2. Measurement Instrument 测量设备

The accuracy of measuring instruments and meters should meet the following requirements:

测量仪器、仪表准确度应满足以下要求：

A. Voltage measuring device 电压测量装置：±0.1%；

B. Current measuring device 电流测量装置：±0.1%；

C. Temperature measuring device 温度测量装置：±0.5°C；

D. Dimension measuring device 尺寸测量装置：±0.01mm；

E. Weight measuring device 重量测量装置：±0.1g.

3.3. Testing Clamp Preparation 测试夹具准备

3.3.1. General Clamp 普通夹具

The single cell needs to be clamped with steel splints (thickness: greater than or equal to 12 mm). The steel splints need to cover the large surface of the cell. The splints are fixed with 10 M6 bolts. All sides of the splints need to cover to be covered with insulating film. Fixtures as shown below:

单体电池需采用钢夹板（厚度：≥10mm）固定，钢板需要覆盖住电池大面，两个钢板之间采用 4 个 M6 螺栓固定，且夹具各个面均需用绝缘膜包覆，绝缘膜厚度为 0.1mm，夹具工装如下图 1 所示：

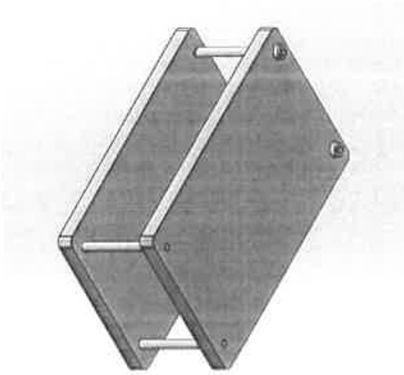

图 1 电池夹具示意图
Fig.1 Schematic diagram of cell clamp

3.3.2. Swelling force Clamp 膨胀力夹具

The single cell needs to be fixed by pressing plate (material: stainless steel 304, thickness: 20mm outside and 15mm inside). The pressing plate needs to cover the large surface of the cell. Four guide pillars are used between the pressing plates to fix, and each side of the pressing plate needs to be covered with insulating film. The thickness of the insulating film is not less than 0.1mm, as shown in Figure 2.

单体电池需采用压板（材质：不锈钢 304，厚度：外侧 20mm、内侧 16.5mm）固定，压板需要覆盖住电池大面，压板之间采用 4 个导柱固定，且压板各个面均需用绝缘膜包覆，绝缘膜厚度不小于 0.1mm，如图 2。

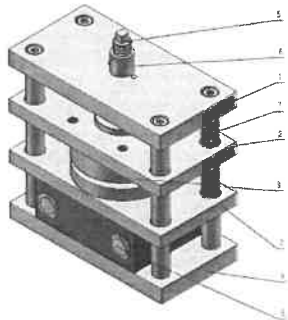

图 2 膨胀力夹具示意图
Fig.2 Schematic diagram of expansion force fixture

3.4. Testing Clamp Installation 测试夹具安装

Place the cell (~20%~30%SOC) covered with blue film (material: PET, thickness 0.1mm) and top film (material:

PC, thickness 0.5mm) in the middle of the clamp, and the initial compression force is (300 kgf±20kgf) as shown in Figure 3 and Figure 4.

将包覆有蓝膜（材质：PET，厚度 0.1mm）和顶贴片（材质：PC，厚度 0.5mm）的电池（~20%~30%SOC）置于夹具中间，每个螺栓初始预紧力为 300kgf±20kgf，如图 3 和图 4。

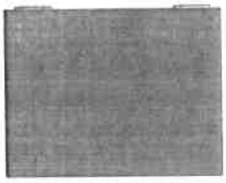

Fig.3 Schematic diagram of cell coating
图 3 电池包膜示意图

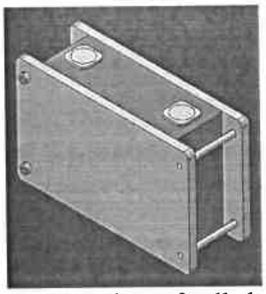

Fig.4 Side view of cell shaft
图 4 电池上夹具图

3.5. Standard Charge 标准充电方式

At ambient temperature of 25°C±2°C, the cell is charged to 3.65V with constant current of 1/3C, then charges at constant voltage of 3.65V until the current decreases to 0.05C, and rest the cell for 30 min.

标准充电是在环境温度 25°C±2°C的条件下，对电池以 1/3C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C。搁置 30min。

3.6. Standard Discharge 标准放电方式

At ambient temperature of 25°C±2°C, the cell is discharged to 2.5V with constant current of 1/3C, and rest the cell for 30 min.

标准放电是在环境温度 25°C±2°C的条件下，电池以 1/3C 的电流恒流放电，放电至电压达到 2.5V 截止。搁置 30min。

3.7. Capacity Calibration and Energy Calibration 容量标定和能量标定

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Then discharge according to chapter 3.6, rest the cell for 30 min. Repeat the above steps 3 times, and the average discharge capacity of the 3 times is the 1/3C discharge capacity, which is recorded as C_0, the average discharge energy of the 3 times is the 1/3C discharge energy, which is recorded as E_0.

容量标定是在环境温度 25°C±2°C的条件下，对电池按照 3.5 标准充电方式进行充电，然后按照 3.6 标准放电进行放电。搁置 30min。将标准充电方式和标准放电方式重复 3 次，3 次的平均放电容量即为 1/3C 放电容量，记录放电容量为标定容量 C_0，3 次的平均放电能量即为 1/3C 放电能量，记录放电能量为标定能量 E_0。

3.8. Test Methods 测试方法

3.8.1. Dimension 尺寸

Testing Instrument: Automatic wrapping machine

试验设备：自动包膜机

Test Method:

试验方法：

a) Thickness, length and height of the delivery cell are measured by automatic wrapping machine;

使用包膜机自动测试出货态电池高度，厚度和宽度；

b) Test conditions: 300kgf ± 20kgf.

测试条件：300kgf ± 20kgf 下测试。

3.8.2. Weight 重量

Test Instrument: electronic scale;

试验设备：电子秤；

Test Method: weight of the cell is measured by electronic.

试验方法：使用电子秤测量电池的重量。

3.8.3. Electrical Performance 电性能

3.8.3.1. 1/3C Discharge Capacity and Energy 1/3C 放电容量和能量

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Then discharge according to chapter 3.6 and rest the cell for 30 min. Record the discharge capacity and discharge energy. Repeat the charging method and discharge method 3 times, and the average discharge capacity of the 3 times is the 1/3C discharge capacity, which is recorded as C_1, the average discharge energy of the 3 times is the 1/3C discharge energy, which is recorded as E_1.

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后按照标准放电方式（3.6）放电，记录放电容量和放电能量。将标准充电方式和标准放电方式重复 3 次，3 次的平均放电容量即为 1/3C 放电容量 C_1，3 次的平均放电能量即为 1/3C 放电能量 E_1。

3.8.3.2. 1C Discharge Capacity and Energy 1C 放电容量和能量

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Then discharge to 2.5V with constant of 1C and rest the cell for 30 min. Record the discharge capacity and discharge energy. Repeat the charging method and discharge method 3 times, and the average discharge capacity of the 3 times is the 1C discharge capacity, which is recorded as C_2, the average discharge energy of the 3 times is the 1C discharge energy, which is recorded as E_2.

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后以 1C 的电流恒流放电至 2.5V，搁置 30min，记录放电容量和放电能量。以上充放电重复 3 次，3 次的平均放电容量即为 1C 放电容量 C_2，3 次的平均放电能量即为 1C 放电能量 E_2。

3.8.3.3. -20°C Capacity Retention Rate -20°C容量保持率

Capacity calibration is carried out according to chapter 3.7. At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. After that, rest the cell at -20°C±2°C for 8 h, and discharge it to 2.0 V with constant current of 1/3C under the environment of -20°C±2°C. Discharge capacity is recorded as C_3, and C_3/C_0 is the capacity retention rate at -20°C.

对电池按照 3.7 的方法进行容量标定。在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后在-20°C±2°C的环境下搁置 8h，在-20°C±2°C的环境下用 1/3C 的电流恒流放电至 2.0V，记录放电容量 C_3，C_3/C_0 即为-20°C容量保持率。

3.8.3.4. 0°C Capacity Retention Rate 0°C容量保持率

Capacity calibration is carried out according to chapter 3.7. At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. After that, rest the cell at 0°C±2°C for 8 h, and discharge it to 2.0 V with constant current of 1/3C under the environment of 0°C±2°C. Discharge capacity is recorded as C_4, and C_4/C_0 is the capacity retention rate at 0°C.

在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定）。在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后在 0°C±2°C的环境下搁置 8h，在 0°C±2°C的环境下用 1/3C 的电流恒流放电至 2.0V，记录放电能量 C_4，C_4/C_0 即为 0°C容量保持率。

3.8.3.5. 35°C Capacity Retention Rate 35°C容量保持率

Capacity calibration is carried out according to chapter 3.7. At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. After that, rest the cell at 35°C±2°C for 8 h, and discharge it to 2.5 V with constant current of 1/3C under the environment of 35°C±2°C. Discharge capacity is recorded as C_5, and C_5/C_0 is the capacity retention rate at 45°C.

在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定）。在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后在 35°C±2°C的环境下搁置 2h，在 35°C±2°C的环境下用 1/3C 的电流恒流放电至 2.5V，记录放电容量 C_5，C_5/C_0 即为 35°C容量保持率。

3.8.3.6. Internal Resistance 内阻

a) ACR: When the SOC is 20%~30% at ambient temperatures, test the cell with a frequency of AC 1 kHz.

b) DCR: Capacity calibration is carried out according to chapter 3.7. After that, the cell is charged according to chapter 3.5, and discharge with constant current of $1/3C_1$ for 90 min afterwards (adjust the SOC to 50%). Then rest for 1h, and record the voltage V_1 at the end of the period. Put a 10 s discharge pulse current of 1C and record the voltage V_2 at the end of the pulse, and calculate the DCR. DCR=(V_1-V_2) *1000/106 (mΩ).

a) 交流内阻（ACR）：在环境温度 25°C±2°C的条件下，20%~30%SOC 的电池采用 AC 1kHz 的频率进行测试。

b) 直流内阻（DCR）：在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定），然后按照标准充电方式（3.5）充电，然后以 $1/3C_1$ 的电流恒流放电 90min（调整 SOC 为 50%）搁置 1h，记录搁

置末期电压 V_1，然后用 106A 的电流恒流放电 10sec，记录放电末期电压 V_2，计算 DCR，DCR=（V_1-V_2）*1000/106 mΩ。

3.8.3.7. 25°C Fast Charge Cycle 25°C快充循环

Before the test, prepare the fixture according to chapter 3.3. When the SOC is 20%~30% at ambient temperature, install the test fixture according to the method of chapter 3.4.

测试前按照 3.3 进行夹具准备，在常温下 20%~30%SOC 时然后按照 3.4 的方法安装测试夹具。

a) Pre-cycle initial capacity test: test the capacity according to chapter 3.7. and record the initial capacity as C_6.

b) First, at ambient temperature of 25°C±2°C, the cell is charge according to Chapter 4.1.2. Rest for 30 min.

c) At ambient temperature of 25°C±2°C, the cell is charged to 2.5V with constant current of $1C_6$ and rest for 30 min;

d) Terminate after 50 cycles according to steps a ~ c.

Capacity test after cycle: at ambient temperature of 25°C±2°C, discharge the cell to 2.5V with constant current of $1/3C_6$. Rest for 30min, then charging it to 3.65V with constant current of $1/3C_6$, and switch to constant voltage charging until the cut-off current is $0.05C_6$. Rest for 30 min, then discharge the cell according to chapter 3.6, and record the discharge capacity C_7. The capacity retention rate = C_7/C_6*100%.

Remarks: The test current needs to be adjusted according to the current capacity every 50 cycles.

a) 循环前初始容量测试：对电池按照 3.7 的方法进行容量测试，记录初始容量 C_6。

b) 在 25°C±2°C的环境温度下对电池以 4.1.2 方法进行充电，搁置 30min；

在 32 容量测试：在 25°C±2°C的环境温度下对电池以 1/3 C_6 的电流恒流放电至 2.5V，搁置 30min，然后按照 1/3 C_6 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 $0.05C_6$ 搁置 30min，然后按照标准放电方式（3.6）放电，记录放电容量 C_7，容量保持率=C_7/C_6×100%。

备注：每循环 50 周需要依据当前容量，调整测试电流。

3.8.3.8. 35°C Fast Charge Cycle 35°C快充循环

Before the test, prepare the fixture according to chapter 3.3. When the SOC is 20%~30% at ambient temperature, install the test fixture according to the method of chapter 3.4.

测试前按照 3.3 进行夹具准备，在常温下 20%~30%SOC 时然后按照 3.4 的方法安装测试夹具。

a) Pre-cycle initial capacity test: test the capacity according to chapter 3.7. and record the initial capacity as C_8.

b) First, at ambient temperature of 35°C±2°C, the cell is charge according to Chapter 4.1.2. Rest for 30 min.

c) At ambient temperature of 35°C±2°C, the cell is charged to 2.5V with constant current of $1C_8$ and rest for 30 min;

d) Terminate after 50 cycles according to steps a ~ c.

Capacity test after cycle: at ambient temperature of 25°C±2°C, discharge the cell to 2.5V with constant current of $1/3C_8$. Rest for 30min, then charging it to 3.65V with constant current of $1/3C_8$, and switch to constant voltage charging until the cut-off current is $0.05C_8$. Rest for 30 min, then discharge the cell according to chapter 3.6, and record the discharge capacity C_9. The capacity retention rate = C_9/C_8*100%.

Remarks: The test current needs to be adjusted according to the current capacity every 50 cycles.

测试前上夹具：在常温下 20%~30%SOC 时然后按照 3.4 的方法安装测试夹具。

循环前能量/容量测试：

a) 在 25°C±2°C的环境温度下对电池以 1/3C 的电流恒流放电至 2.5V，搁置 30min，然后按照 1/3C 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05C 搁置 30min。，然后按照标准放电方式（3.6）放电，记录放电容量 C_8；

b) 在 35°C±2°C的环境温度下对电池以 4.1.2 方法进行充电，搁置 30min；

c) 在 35°C±2°C的环境温度下对电池以 $1C_8$ 的电流恒流放电至 2.5V，搁置 30min；

d) 重复 a-d 循环 50 周。

循环后容量/能量测试：在 25°C±2°C的环境温度下对电池以 1/3 C_8 的电流恒流放电至 2.5V，搁置 30min，然后按照 1/3 C_8 的电流恒流充电至 3.65V，然后在 3.65V 下转恒压充电，直至充电电流小于等于 0.05 C_8，搁置 30min。，然后按照标准放电方式（3.6）放电，记录放电容量 C_9，容量保持率=$C_9/C_8\times100\%$。

备注：每循环 50 周需要依据当前容量，调整测试电流。

3.8.3.9. 25°C Storage 25°C存储

Capacity calibration is carried out according to chapter 3.7. Record the capacity C_{10}. At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Rest for 28 days afterwards at ambient temperature of 25°C±2°C. At ambient temperature of 25°C±2°C, discharging the cell according to chapter 3.6. Record the discharge capacity C_{11}. Then the cell is charged according to chapter 3.5, and discharging the cell according to chapter 3.6. Record the discharge capacity C_{12}. Capacity retention = $C_{11}/C_{10}\times100\%$, Capacity recovery = $C_{12}/C_{10}\times100\%$.

在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定），标定后的容量为 C_{10}，然后按照标准充电方式（3.5）充电，然后在环境温度 25°C±2°C的条件下搁置 28 天，然后在环境温度 25°C±2°C的条件下按照标准放电方式（3.6）放电（记录放电容量 C_{11}），然后按照准充电方式（3.5）充电后用标准放电方式（3.6）放电（记录放电容量 C_{12}）。容量保持率=$C_{11}/C_{10}\times100\%$，容量恢复率=$C_{12}/C_{10}\times100\%$。

3.8.3.10. 45°C Storage 45°C存储

Capacity calibration is carried out according to chapter 3.7. Record the capacity C_{13}. At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Rest for 28 days afterwards at ambient temperature of 45°C±2°C. At ambient temperature of 45°C±2°C, discharging the cell according to chapter 3.6. Record the discharge capacity C_{14}. Then the cell is charged according to chapter 3.5, and discharging the cell according to chapter 3.6. Record the discharge capacity C_{15}. Capacity retention = $C_{14}/C_{13}\times100\%$, Capacity recovery = $C_{15}/C_{13}\times100\%$.

在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定），标定后的容量为 C_{13}，然后按照标准充电方式（3.5）充电，然后在环境温度 45°C±2°C的条件下搁置 28 天，然后在环境温度 25°C±2°C的条件下搁置 4h，然后按照标准放电方式（3.6）放电（记录放电容量 C_{14}），然后按照准充电方式（3.5）充电后用标准放电方式（3.6）放电（记录放电容量 C_{15}）。容量保持率=$C_{14}/C_{13}\times100\%$，容量恢复率=$C_{15}/C_{13}\times100\%$。

3.8.4. Safety Performance 安全性能

3.8.4.1. Over Discharge 过放电

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Then the cell is discharged with constant current of 1C for 90 min at the ambient temperature of the safety test. Observe for 1h. (Refer to GB 38031-2020 electric vehicles traction battery safety requirements)

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后在安全试验环境温度 25°C±5°C下电池以 1C 恒流放电 90 min。观察 1h。（参考 GB 38031-2020 电动汽车用蓄电池安全要求）

3.8.4.2. Over Charge 过充电

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Then the cell is charged to 4.015V, or 115% SOC with constant current of 1C at the ambient temperature of the safety test. Observe for 1h. (Refer to GB 38031-2020 electric vehicles traction battery safety requirements)

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电，然后按照 3.4 的方法安装测试夹具。在安全试验环境温度下电池以 1C 恒流充电至 4.015V 或 115%SOC 后，停止充电。观察 1h。（参考 GB 38031-2020 电动汽车用蓄电池安全要求）

3.8.4.3. Extrusion 挤压

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Test under the following conditions at a safety test environment temperature of 25°C±2°C:

a) Extrusion direction: apply pressure perpendicular to the direction of the cell plate, or the same direction that the cell is most susceptible to extrude in the layout of the whole vehicle;

b) The form of the extruded plate: a semi-cylinder with a radius of 75mm, the length (L) of the semi-cylinder is greater than the size of the cell being extruded;

c) Extrusion speed: not more than 2 mm/s;

d) Extrusion degree: stop extruding after the voltage reaches 0 V or the deformation reaches 15% or the extruding force reaches 100 KN or 1000 times the weight of the test object;

e) Keep it for 10 min. Observe for 1 h. (Refer to GB 38031-2020 electric vehicles traction battery safety requirements)

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电。在安全试验环境温度 25°C±5°C下按照如下条件进行试验：

a) 挤压方向：垂直于电池单体极板方向施压，或与电池单体在整车布局上最容易受到挤压的方向相同；

b) 挤压板形式：半径 75mm 的半圆柱体，半圆柱体的长度（L）大于被挤压电池单体的尺寸；

c) 挤压速度：不大于 2mm/s；

d) 挤压程度：电压达到 0V 或变形量达到 15%或挤压力达到 100KN 或 1000 倍试验对象重量后停止挤压；

e) 保持 10min。观察 1h。（参考 GB 38031-2020 电动汽车用蓄电池安全要求）

3.8.4.4. External Short-circuit 外部短路

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. The positive and negative terminals of the cell are short-circuited externally for 10 min under the environmental temperature of the safety test, and the resistance of the external circuit should be less than 5 mΩ. Observe for 1 h. (Refer to GB 38031-2020 electric vehicles traction battery safety requirements)

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电。在安全试验环境温度 25°C±5°C下将电池正、负极经外部短路 10min，外部线路电阻值应小于 5mΩ,观察 1h。(参考 GB 38031-2020 电动汽车用蓄电池安全要求)

3.8.4.5. Heating(130°C) 加热（130°C）

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Put the cell into the temperature chamber, and the temperature chamber will rise from room temperature to 130°C±2°C at a rate of 5°C/min, and keep this temperature for 30 min before stopping heating. Observe for 1 h. (Refer to GB 38031-2020 electric vehicles traction battery safety requirements)

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电。将电池放入温度箱，温度箱按照 5°C/min 的速率由室温升至 130°C±2°C，并保持此温度 30min 后停止加热。观察 1h。(参考 GB 38031-2020 电动汽车用蓄电池安全要求)

3.8.4.6. Temperature Cycling 温度循环

At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5. Put the cell into the temperature chamber, and adjust the temperature chamber according to the following table and figure, and cycles for 5 times. (Refer to GB 38031-2020 electric vehicles traction battery safety requirements)

在环境温度 25°C±2°C的条件下，对电池按照标准充电方式（3.5）充满电。将电池放入温度箱中，温度箱按照下表进行调节，循环次数 5 次。参考 GB 38031-2020 电动汽车用蓄电池安全要求)

Table 5 Temperature cycle corresponding parameter

表 5 温度循环对应参数

Temperature (°C) 温度（°C）	Time Increment (min) 时间增量（min）	Time Accumulation (min) 累计时间（min）	Temperature Change Rate (°C/min) 温度变化率（°C/min）
25	0	0	0
-40	60	60	13/12
-40	90	150	0
25	60	210	13/12
85	90	300	2/3
85	110	410	0
25	70	480	6/7

4. Parameters Recommendation for BMS BMS 设计参数建议

The following data is the reference performance data of A27-V1 battery for reference in BMS design. The actual use is subject to the use mode and conditions agreed by both parties.

以下数据为 A27-V1 电池参考性能数据，供 BMS 设计时参考使用，实际使用以双方约定的使用方式和条件为准。

4.1. Electrical performance data 电性能数据

4.1.1. SOC~OCV

Capacity calibration is carried out according to chapter 3.7. At ambient temperature of 25°C±2°C, the cell is charged according to chapter 3.5, and discharge it with consistent current for 1/3C, each discharge capacity is 10% × C_0. Rest for 180 min, repeat discharge the cell 10 times. Record voltage after rest.

Capacity calibration is carried out according to chapter 3.7. At ambient temperature of 25°C±2°C, and charge it with consistent current for 1/3C, each charge capacity is 10% × C_0. Rest for 180 min, repeat discharge the cell 10 times. Record voltage after rest.

电池在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定），然后按照标准充电方式（3.5）充电，然后以 1/3C 的电流恒流放电，每次放电容量为 10%×C_0，搁置 180min，重复放电 10 次，记录每次搁置后的电压，作为放电态下 SOC 所对应的 OCV。

电池在环境温度 25°C±2°C的条件下，对电池进行容量标定（3.7 容量标定和能量标定），然后以 1/3C 的电流恒流充电，每次充电容量为 10%×C_0，搁置 180min，重复充电 10 次，记录每次搁置后的电压，作为充电态下 SOC 所对应的 OCV。

Table 6 Charge and discharge SOC/OCV data

表 6 充放电 SOC/OCV 数据

温度	OCV (mV)											
	SOC%	100	90	80	70	60	50	40	30	20	10	0
25°C	放电	3437	3331	3330	3329	3305	3292	3290	3283	3249	3205	2815
	充电	3456	3340	3341	3341	3319	3308	3305	3303	3277	3223	2836

4.1.2. Recommended charging 推荐充电

Conventional charge (10°C~45°C): charging to 3.65V with constant current of 1/3C, and then switch to constant voltage charging at 3.65V until the charging current reaches 0.05C.

Fast charge (25°C,10%~80%SOC): Standard charging mode is shown in the following table.

常规充电（10°C~45°C）：1/3C 恒流恒压充电至 3.65V，0.05C 截止。

快速充电（25°C,10%~80%SOC）：快充策略如下表。

Table 7 Fast charge mode

表 7 快充策略

charging mode 充电方式	SOC Range SOC 区间	25 °C Charge Rate 25°C 充电倍率	35 °C Charge Rate 35°C 充电倍率
Constant current charging 恒流充电	0%~5% SOC	1.0C	1.0C
Constant current charging 恒流充电	5%~10% SOC	1.5C	1.5C
Constant current charging 恒流充电	10%~15% SOC	2.0C	2.0C
	15%~20% SOC	2.5C	2.5C

	20%~25% SOC	2.5C	2.5C
	25%~30% SOC	2.5C	2.5C
	30%~35% SOC	2.5C	2.5C
Constant current charging 恒流充电	35%~40% SOC	2.0C	2.0C
Constant current charging 恒流充电	40%~45% SOC	2.0C	2.0C
Constant current charging 恒流充电	45%~50% SOC	2.0C	2.0C
Constant current charging 恒流充电	50%~55% SOC	2.0C	2.0C
Constant current charging 恒流充电	55%~60% SOC	1.5C	1.5C
Constant current charging 恒流充电	60%~65% SOC	1.5C	1.5C
Constant current charging 恒流充电	65%~70% SOC	1.5C	1.5C
Constant current charging 恒流充电	70%~75% SOC	1.5C	1.5C
Constant current charging 恒流充电	75%~80% SOC	1.0C	1.0C
Constant current charging 恒流充电	80%~85% SOC	0.85C	0.85C
Constant current charging 恒流充电	85%~90% SOC	0.55C	0.55C
Constant current charging 恒流充电	90%~95% SOC	0.5C	0.5C
Constant current charging 恒流充电	95%~98% SOC	0.33C	0.33C
Constant current charging 恒流充电	98%~99% SOC	0.2C	0.2C
Constant current charging 恒流充电	99%~100% SOC	0.2 C	0.2 C

4.1.3. Pulsing Charge and Pulsing Feedback 脉冲放电和充电功率

Table 8 Pulsing Charge and Pulsing Feedback
表 8 脉冲放电和充电功率

T (°C) 温度	Mode 方式	Time (sec) 时间(sec)	Maximum power (W) 最大功率（W）							
			90% SOC	I_{max} (A)	80%SOC	I_{max} (A)	50% SOC	I_{max} (A)	20% SOC	I_{max} (A)
45	Discharge 放电	10	2500	1000	2500	1000	2500	1000	2081	832
		60	2450	980	2450	980	2300	920	1650	660
25		10	2500	1000	2500	1000	2313	925	1948	779
		60	2413	965	2363	945	2038	815	1163	465
0		10	2325	930	2250	900	1950	780	1241	496
		60	1480	740	1346	673	1004	502	535	268
-10		10	1012	506	980	490	846	423	540	270
		60	977	489	886	443	557	279	291	145
45	Charge 充电	10	1285	385	1376	412	1621	490	2029	620
		60	598	179	819	246	1022	307	1343	339
25		10	835	250	929	278	1141	345	1405	410
		60	476	134	501	141	635	179	753	212

0		10	247	74	251	75	281	85	370	113
		60	117	33	128	36	195	55	241	68
-10		10	74	22	77	23	93	28	184	56
		60	36	10	39	11	64	18	96	27

4.1.4. DCR

Table 9 DCR

表 9 DCR

T (°C)	1C DCR (mΩ)				
	Time (sec)	90% SOC	50% SOC	30% SOC	20% SOC
45	10	0.558	0.593	0.637	0.665
25	10	0.831	0.922	1.019	1.097
0	10	2.586	2.867	3.146	3.400
-20	10	6.324	6.927	7.502	8.037

4.1.5. Charging capacity at different temperatures 不同温度充电容量

After standard discharge, the battery shall be stored for 4h/8h at the corresponding temperature in the following table, and then charged to 3.65V with constant current of 1/3C. The tested capacity is the charging capacity at the corresponding temperature and current.

标准放电后的电池，在如下表格中对应的温度下搁置 4h/8h，然后以 1/3C 的电流恒流充电至 3.65V。测试所得容量为对应温度、对应电流下的充电容量。

Table 10 charge capacity

表 10 充电容量

Charge Rate 充电倍率	T (°C) 温度（°C）	Charge Capacity (Ah) 充电容量（Ah）
1/3C	45	106
	25	106
	10	103.2

4.1.6. Discharging capacity at different temperatures 不同温度放电容量

After standard charge, the battery shall be stored for 4h/8h at the corresponding temperature in the following table, and then discharged to 2.5V/2.0V with constant current of 1/3C. The tested capacity is the discharging capacity at the corresponding temperature and current.

标准充电后的电池，在如下表格中对应的温度下搁置 4h/8h，然后分别以 1/3C 的电流恒流放电至 2.5V/2.0V，测试所得容量为对应温度、对应电流下的放电容量。

Table 11 discharge capacity

表 11 放电容量

Discharge Rate 放电倍率	T (°C) 温度（°C）	Discharge Capacity (Ah) 放电容量（Ah）
1/3C	45	106
	25	106
	0	93
	-20	74.4

4.2. Battery safe operation limit 电池安全操作限制

4.2.1. Current Limits 电流限制

4.2.1.1. Discharge operation current limit 放电操作电流限制

Table 12 Discharge operation current limit
表 12 放电操作电流限制

T (°C) 温度 (°C)	Discharge operation current limit 放电操作电流限制			
	Peak Imax (A) 峰值 Imax (A)	Continuous current limit 持续电流限制		
		Time (sec) 时间 (sec)	Continuous max current 持续最大电流 Imax	
			Continuous I max(A) 持续 Imax(A)	Use standard allowed 允许的使用标准
56	0	-	0	-
55	1000	60	980	100%
50	1000	60	980	100%
40	1000	60	980	100%
35	1000	60	980	100%
30	1000	60	980	100%
25	1000	60	980	100%
20	990	60	950	100%
15	990	60	900	100%
10	980	60	900	100%
5	950	60	850	100%
0	950	60	800	100%
-5	600	60	580	100%
-10	520	60	500	100%
-15	450	60	400	100%
-20	360	60	345	100%
-25	200	60	120	100%
-30	96	60	77	100%
-31	0	-	0	-

Remarks: The duration of peak current is 10s

注：峰值电流的耐受时间为 10s。

4.2.1.2. Charge operation current limit 充电操作电流限制

Table 13 Charge operation current limit
表 13 充电操作电流限制

T (°C) 温度 (°C)	Charge operation current limit 充电操作电流限制		
	Peak Imax (A) 峰值 Imax (A)	持续电流限制	
		Time (sec) 时间 (sec)	Continuous max current 持续最大电流 Imax

			Continuous I max(A) 持续 Imax(A)	Use standard allowed 允许的使用标准
56	0	-	0	-
55	185	60	148	100%
50	321	60	256	100%
45	800	60	640	100%
40	800	60	640	100%
35	750	60	600	100%
30	680	60	544	100%
25	679	60	543	100%
20	600	60	480	100%
15	570	60	456	100%
10	550	60	440	100%
5	390	60	312	100%
0	218	60	174	100%
-5	165	60	132	100%
-10	97	60	77	100%
-15	50	60	40	100%
-20	46	60	36	100%
-30	0	-	0	-

Remarks: The duration of peak current is 10s

注：峰值电流的耐受时间为 10s。

4.2.1.3. Safety current limit 安全电流限制

If the current exceeds $I_{max\text{-}safety}$ within the range of 0msec to 200msec, the cell will not trigger a safety event (EUCA.R hazard level ≤ L3: Battery leakage and electrolyte loss < 50%) However, the cell cannot continue to charge and discharge, and must be replaced.

If the cell is used between the operating current limit and the safety current limit, the cell will seriously accelerate the attenuation, but no safety event will occur.

If no temperature is specified, the safe limit current can be determined by linear interpolation between two adjacent conditions in the following table.

如果在 0msec 到 200msec 范围内电流超过 $I_{max\text{-}safety}$，电池不会触发安全事件（EUCA.R 危险等级≤L3：电池漏液，且电解液损失< 50%），但是该电池不能继续充放电，且必须更换。

如果在操作电流限制与安全电流限制之间使用，电池会严重加速衰减，但不会发生安全事件。

在未指定温度的情况下，可通过下表中两个相邻条件之间的线性插值来确定安全限制电流。

Table 14 Safety limit current parameters

表 14 安全限制电流参数

Temperature (°C) 温度 (°C)	Safety current limit 安全电流限制			
	Discharge 放电		Charge 充电	
	Peak Imax (A) 峰值 Imax (A)	Time (msec) 时间(msec)	Peak Imax (A) 峰值 Imax (A)	Time (msec) 时间(msec)
56	0	0	0	0

55	1100	1000	203	1000
50	1100	1000	353	1000
40	1100	1000	880	1000
35	1100	1000	880	1000
30	1100	1000	825	1000
25	1089	1000	748	1000
20	1089	1000	746	1000
15	1078	1000	660	1000
10	1045	1000	627	1000
5	1045	1000	605	1000
0	660	1000	429	1000
-5	572	1000	239	1000
-10	495	1000	181	1000
-15	396	1000	106	1000
-20	220	1000	55	1000
-25	105	1000	50	1000
-30	0	1000	0	1000
-31	-	-	-	-

4.2.2. Voltage Limits 电压限制

Table 15 Safety limit voltage parameters
表 15 安全限制电压参数

Item 项目	Category 类别	Parameters 参数	Protective Action 保护动作
Charging Voltage 充电电压	Charging Ends 充电终止	3.65V	When the cell voltage reaches 3.65V, stop charging 当电池的电压达到 3.65V 时终止充电
	First Over-Charging Protection 第一级过充保护	3.70V	When the cell voltage reaches 3.70V, Forewarning 预报警
	Second Over-Charging Protection 第二级过充保护	3.85V	When the cell voltage reaches 3.85V, Forced stop 强制停止
放电电压	Discharging Ends 放电终止	2.5V 2.0V	Temperature T > 20°C. Stop charging when the cell discharge reaches 2.5V Temperature T ≤ 20°C. Stop charging when the cell discharge reaches 2.0V 温度 T > 20°C 电池放电达到 2.5V 时停止充电 温度 T ≤ 20°C 电池放电达到 2.0V 时停止充电
	First Over-Discharging Protection 第一级过放保护	2.00V 1.90V	Temperature T > 20°C. Forewarning when the cell discharge reaches 2.0V Temperature T ≤ 20°C. Forewarning when the cell discharge reaches 1.90V 温度 T > 20°C 电池放电达到 2.0V 时预报警 温度 T ≤ 20°C 电池放电达到 1.90V 时预报警

	Second Over-Discharging Protection 第二级过放保护	1.85V 1.75V	Temperature T > 20°C. Forced stop when the cell discharge reaches 1.85V Temperature T ≤ 20°C. Forced stop when the cell discharge reaches 1.75V 温度 T > 20°C 电池放电达到 1.85V 时强制停止 温度 T ≤ 20°C 电池放电达到 1.75V 时强制停止

4.2.3. Temperature limits 温度限制

Table 16 Safety limit temperature parameters
表 16 安全限制温度参数

Item 项目	Value 数值	Remarks 备注
Recommended Operating Temperature Range 推荐操作温度范围	10°C~45°C	Recommend cell usage temperature range 推荐使用电池的温度范围。
Maximum operating temperature 最高操作温度	55°C	If the cell temperature exceeds the maximum operating temperature, the power needs to be reduced to 0. 如果电池使用温度超过最高操作温度，功率需要降为 0。
Minimum operating temperature 最低操作温度	-30°C	If the cell temperature exceeds the minimum operating temperature, the power needs to be reduced to 0. 如果电池使用温度超过最低操作温度，功率需要降为 0。
Maximum safe temperature 最高安全温度	65°C	If the cell temperature exceeds the maximum safe temperature, it will cause irreversible and permanent damage to the cell, and the user should not use it higher than the maximum safe temperature. 如果电池使用温度超过最高安全温度，将会造成电池不可逆的永久性损坏，用户使用时不得高于最高安全温度。
Minimum safe temperature 最低安全温度	-35°C	If the cell temperature exceeds the minimum safe temperature, it will cause irreversible and permanent damage to the cell, and the user should not use it lower than the minimum safe temperature. 如果电池使用温度超过最低安全温度，将会造成电池不可逆的永久性损坏，用户使用时不得低于最低安全温度。

Remarks 备注：

a) Avoid charging the cell at low temperatures (including but not limited to standard charge, quick charge, emergency charge and regenerative charge) prohibited by this specification, otherwise unexpected capacity reduction may occur. The cell management system should be controlled according to minimum charging and regenerative charging temperatures. Charging at temperatures lower than specified in this specification is prohibited, otherwise, EVE will not bear all relevant responsibilities such as quality assurance liability and loss compensation caused thereby.

b) The heat dissipation of battery should be fully considered in the design of battery pack, EVE is not responsible for the quality assurance caused by overheating due to the heat dissipation design of battery pack.

a) 电池避免在本规格书禁止的低温条件下充电（包括但不限于标准充电，快充，紧急情况充电和再生充电等），否则可能出现意外的容量降低现象。电池管理系统应依照最小的充电和再生充电温度进行控制。禁止在低于本规格书规定的温度条件下充电，否则，EVE 不承担质量保证责任及由此引起的损失赔偿等一切相关责任。

b) 电池包设计中应充分考虑电池的散热问题，由于电池包散热设计问题导致的电池或电池过热损坏，EVE 不承担质量保证责任。

5. Parameters Recommendation for Module Design 模组设计参数建议

5.1. Cell Directions 电池方向

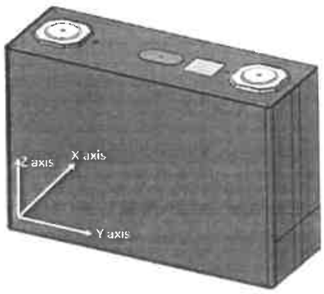

Fig.5 Schematic diagram of cell direction
图 5 电池方向示意图

5.2. Cell Compression Force 电池压缩力

Test Conditions 测试条件：

a) Compression area 压缩面积：148.6mm×115.7mm(L×H)

b) Compression speed 压缩速度：0.02mm/sec

c) Compression direction 压缩方向：X direction X 方向

d) Cell SOC 电池 SOC：20%~30%SOC

Table 17 Cell compression force limit parameters
表 17 电池压缩力限制参数

Observation 现象	Compression Force 压缩力
Internal defects 内部产生缺陷	30kN
Leakage 漏液	>100kN

It can be seen from the above table, that the compression force of the cell cannot exceed 30 kN, otherwise the cell may be damaged.

从上表可知，电池承受的压缩力不能超过 30 kN，否则可能电池会受到损害。

5.3. Cell Expansion Force 电池膨胀力

5.3.1. Testing Conditions 测试条件：

Initial force of battery 电池初始预紧力：< 3000Ne

Charging and discharging conditions 充/放电条件：

Charging: Charge according to Chapter 4.1.2. Rest for 30 min.

Discharging: Discharge to 2.5V with constant current of 1C. Rest for 30 min.

Terminate after 2000 cycles. Record the battery expansion force before and after the cycle.

充电：按 4.1.2 推荐快速充电策略充电，搁置 30min。

放电：1C 恒流放电至 2.5V，搁置 30min。

按照充电&放电条件，循环 2000 周，记录循环前后的电池膨胀力。

5.3.2. Test result 测试结果

Table 18 Cell expansion force parameter

表 18 电池膨胀力参数

Expansion Force 膨胀力	BOL	≤3000N
	80% SOH	≤20000N
	75%SOH	≤25000N

5.4. Thermal parameters 热学参数

Test methods: Refer to GB/T 10295-2008、ASTM E1269-2011

测试方法：参考 GB/T 10295-2008、ASTM E1269-2011

Table 19 Thermal parameters

表 19 热学参数

Mean value of thermal conductivity 导热系数均值	Thermal conductivity 导热系数（W/mK)	
	X	Y/Z
	1.5~2W/mK	16~17 W/mK
Average thermal capacity 热容均值	Thermal capacity 热容(kJ/(kg·K))	
	0.9~1.2 kJ/(kg·K)	

5.5. Recommend Temperature Collection Points 推荐温度采集点

When collecting temperature on the cell surface, it is recommended that the temperature collection points to be arranged at the center of the poles and the surface, as shown in the figure below.

对电池表面进行温度采集时，建议温度采集点布置在正极极柱、大面中心及侧面中心处，如下图。

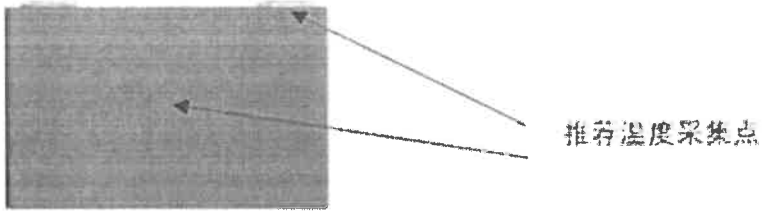

Fig.6 Battery temperature collection point

图 6 电池温度采集点

6. Cell Operation Instructions and Precautions 电池操作说明及注意事项

6.1. Long-term Storage 长期存储

After the battery is charged, it should be used as soon as possible to avoid loss of usable capacity due to self-discharge. If storage is required, the battery needs to be stored in a low SOC state. The recommended storage conditions are 20% ~ 50%SOC, 0 ~ 25°C, ≤60%RH.

电池进行充电后，需尽快使用，以免因自放电而造成可用容量损失。若需要存储，则电池需要在低 SOC 态下进行存储。推荐的存储条件为：20%~50%SOC，0~25°C，≤60%RH。

6.2. Transportation 运输

Battery for shipping should be packed in boxes with the SOC not more than 30% SOC. The severe vibration, impact, extrusion, sun and rain should be prevented during shipping. Applicable methods of transportation include truck, train, ship, airplane, etc.

产品的运输应在不大于 30%SOC 下包装成箱进行。在运输过程中应防止剧烈振动、冲击或挤压、避免日晒雨淋。适用于汽车、火车、轮船、飞机等交通工具运输。

6.3. Operation Precautions 操作说明

- It is forbidden to inversely charge. Correctly connect the positive and negative poles of the battery, and reverse charging is strictly prohibited.
  禁止反向充电。正确连接电池的正负极，严禁反向充电。
- It is forbidden to over-discharge. During the normal use of the battery, in order to prevent over-discharge, the battery should be charged regularly to maintain the voltage above 2.5 V.
  禁止过放电。在电池正常使用过程中，为防止过放电，电池应定期充电，将电压维持在 2.5 V 以上。
- It is strictly forbidden to immerse the battery in water. When it is not in use, it should be placed in a cool and dry environment.
  严禁将电池浸入水中，保存不用时，应放置于阴凉干燥的环境中。
- It is forbidden to use and leave the battery next to heat and high temperature sources, such as fire, heater, etc.
  禁止将电池放在热高温源旁边，如火、加热器等使用和留置。

- Please use a special charger for lithium-ion batteries when charging.
  充电时请选用锂离子电池专用充电器。
- During usage, it is strictly prohibited to reverse the positive and negative terminals of the battery.
  在使用过程中，严禁将电池正负极颠倒。
- Do not throw the battery in the fire or heater.
  禁止将电池丢于火或给电池加热。
- It is forbidden to use metal to directly connect the positive and negative terminals of the battery to short-circuit.
  禁止用金属直接导通电池正负极。
- It is forbidden to transport or store the battery with metal, such as hairpins, necklaces, etc.
  禁止将电池与金属，如发夹、项链等一起运输或贮存。
- It is forbidden to knock or throw, step on, or bend the battery.
  禁止敲击或抛掷、踩踏和弯折电池等。
- It is forbidden to directly solder the battery.
  禁止直接焊接电池。
- It is forbidden to directly pierce the battery with nails or other sharp objects.
  禁止用钉子或其它利器刺穿电池。
- It is forbidden to use or place the battery under high temperature (under hot sunlight), such as in a car under direct sunlight or in a hot day.
  不要使用处于极热环境中的电池，如阳光直射或热天的车内。
- It is forbidden to use it in places with strong static electricity and strong magnetic fields.
  禁止在强静电和强磁场的地方使用。
- If the battery leaks and the electrolyte splashes on the skin, clothes, eyes, mouth, nose, etc., immediately wash the affected area with running water and send to a doctor for treatment immediately, otherwise it will cause serious harm to the human body.
  如果电池漏液，电解液溅入到皮肤、眼睛、口、鼻等部位，应立即用大量清水冲洗，并马上送医治疗，否则会对人体造成严重伤害。
- If the battery emits peculiar smell, heat, discoloration, deformation, or any abnormality during use, storage, or charging, stop using it.
  如果电池出现异味、发热、变色、变形或使用、贮存、充电过程中出现任何异常时不得使用。

6.4. Other 其它

Any matter not mentioned in this specification must be negotiated and determined by both parties.

任何本规格书中未提及的事项，须经双方协商确定。

7. Contact Information 联系方式

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联系地址：湖北省荆门市经济开发区高新区·掇刀区荆南大道 68 号，湖北亿纬动力有限公司

Tel 联系电话：86-0724-6079699

Fax 传真：86-0724-6079688

Website 网址：http://www.evebattery.com

附图：A41-V1 电池图纸

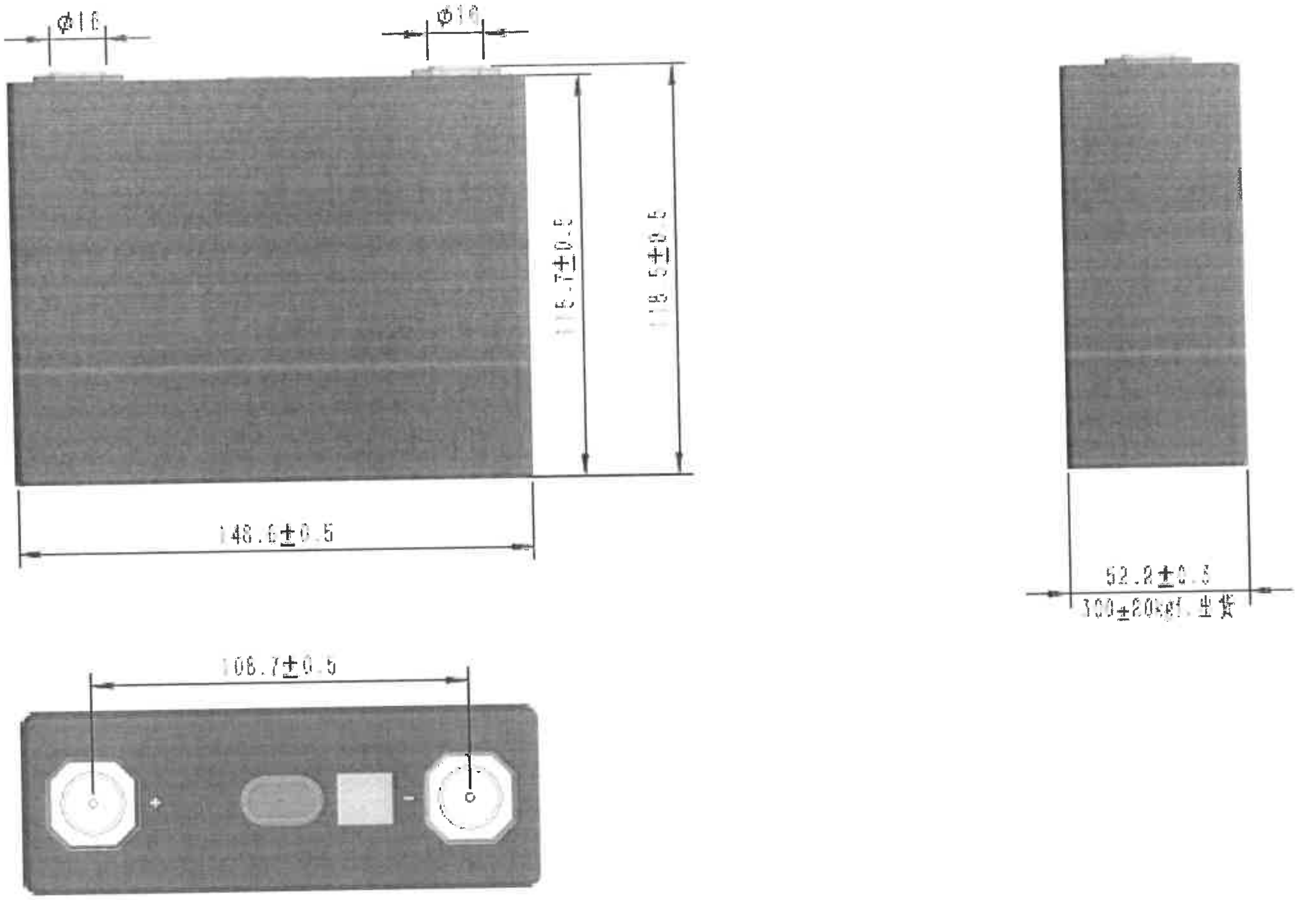

Fig.7 Cell Drawing of A41-V1
图 7 A41-V1 电池